BAHRIA UNIVERSITY, ISLAMABAD

Department of Computer Science



INTROUCTION TO DATA SCIENCE LAB

L A B J O U R N A L # 0 4

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6TH A

SUBMITTED TO

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TASK 1

Write data types of bigsales.csv from above mentioned types

Feature Name	Data type

```
Item_Identifier                 str
Item_Weight                 float64
Item_Fat_Content                str
Item_Visibility             float64
Item_Type                       str
Item_MRP                    float64
Outlet_Identifier               str
Outlet_Establishment_Year     int64
Outlet_Size                     str
Outlet_Location_Type            str
Outlet_Type                     str
Item_Outlet_Sales           float64
dtype: object
```

TASK 2

How to convert the categorical data into numeric data in python by using Pandas?

```
# How to convert the categorical data into numeric data in python by using
Pandas?
import pandas as pd
data = pd.read_csv('Week4/Train.csv')
data['code'] = pd.factorize(data.Item_Fat_Content)[0]
print(data.head())

from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
data['code'] = le.fit_transform(data.Item_Fat_Content)
print(data.head())
```

```
  Item_Identifier  Item_Weight Item_Fat_Content  Item_Visibility  ... Outlet_Location_Type        Outlet_Type Item_Outlet_Sales  code
0           FDA15         9.30          Low Fat         0.016047  ...               Tier 1  Supermarket Type1         3735.1380     0
1           DRC01         5.92          Regular         0.019278  ...               Tier 3  Supermarket Type2          443.4228     1
2           FDN15        17.50          Low Fat         0.016760  ...               Tier 1  Supermarket Type1         2097.2700     0
3           FDX07        19.20          Regular         0.000000  ...               Tier 3      Grocery Store          732.3800     1
4           NCD19         8.93          Low Fat         0.000000  ...               Tier 3  Supermarket Type1          994.7052     0

[5 rows x 13 columns]
  Item_Identifier  Item_Weight Item_Fat_Content  Item_Visibility  ... Outlet_Location_Type        Outlet_Type Item_Outlet_Sales  code
0           FDA15         9.30          Low Fat         0.016047  ...               Tier 1  Supermarket Type1         3735.1380     1
1           DRC01         5.92          Regular         0.019278  ...               Tier 3  Supermarket Type2          443.4228     2
2           FDN15        17.50          Low Fat         0.016760  ...               Tier 1  Supermarket Type1         2097.2700     1
3           FDX07        19.20          Regular         0.000000  ...               Tier 3      Grocery Store          732.3800     2
4           NCD19         8.93          Low Fat         0.000000  ...               Tier 3  Supermarket Type1          994.7052     1

[5 rows x 13 columns]
```

TASK 3

The data set is given (download cars/ vehicles dataset), you have to solve the null or missing values by

all the best possible ways. Handle the null values for the particular columns also.

TASK 4

Bigmart-sales dataset

Retail is another industry which extensively uses analytics to optimize business processes. Tasks like product placement, inventory management, customized offers, product bundling, etc. are being smartly handled using data science techniques. As the name suggests, this data comprises of transaction records of a sales store. The data has 8523 rows of 12 variables

Import bigmart-sales dataset

- display data
- display datatypes of attributes
- filter by item_type=frozen foods
- append 5 records in existing dataset sort in ascending order and display
- plot boxplot and histogram by outlet_type mention no of bins in histogram
- replace item_weight missing values with “1.00”
- replace outlet_size missing values with “Small”
- drop rows whose outlet_type is missing display no of rows after
- plot boxplot and histogram by outlet_type mention no of bins in histogram compare results after.

```
import pandas as pd
import matplotlib.pyplot as plt
data = pd.read_csv("Week4/Train.csv")
print("Data:")
print(data.head())
print("\nData Types:")
print(data.dtypes)
frozen = data[data["Item_Type"] == "Frozen Foods"]
print("\nFrozen Foods:")
print(frozen)
new_records = data.head(5)
data = pd.concat([data, new_records], ignore_index=True)

print("\nAfter Appending 5 Records:")
print(data.tail())
data = data.sort_values(by="Item_Identifier")
print("\nData in Ascending Order:")
print(data.head(10))
# --------------------------------
# BEFORE HANDLING MISSING VALUES
# --------------------------------

# Boxplot by Outlet_Type
data.boxplot(
```

```
    column="Item_Outlet_Sales",
    by="Outlet_Type"
)

plt.title("Boxplot Before")
plt.suptitle("")
plt.xlabel("Outlet Type")
plt.ylabel("Item Outlet Sales")
plt.xticks(rotation=20)
plt.show()


# Histogram by Outlet_Type
data.hist(
    column="Item_Outlet_Sales",
    by="Outlet_Type",
    bins=10
)

plt.suptitle("Histogram Before")
plt.show()

print("\nNumber of bins = 10")


# Check missing values
print("\nMissing Values Before:")
print(data.isnull().sum())


# Replace Item_Weight missing values with 1.00
data["Item_Weight"] = data["Item_Weight"].fillna(1.00)


# Replace Outlet_Size missing values with Small
data["Outlet_Size"] = data["Outlet_Size"].fillna("Small")

# Drop rows where Outlet_Type is missing
data = data.dropna(subset=["Outlet_Type"])

# Display number of rows after
print("\nNumber of Rows After:")
print(data.shape[0])

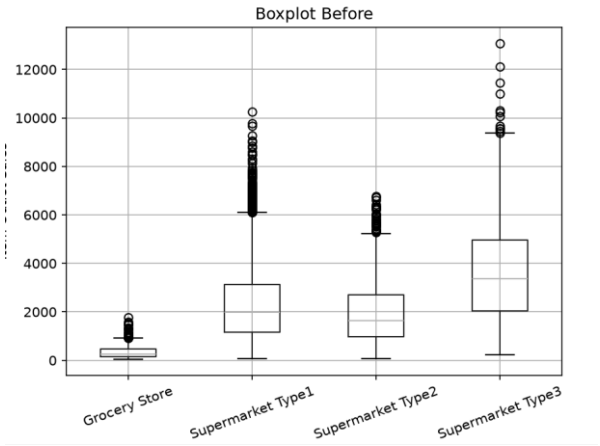

# Check missing values after
print("\nMissing Values After:")
print(data.isnull().sum())

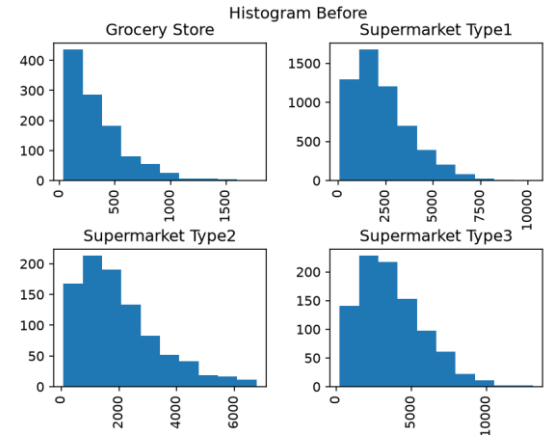
# AFTER HANDLING MISSING VALUES

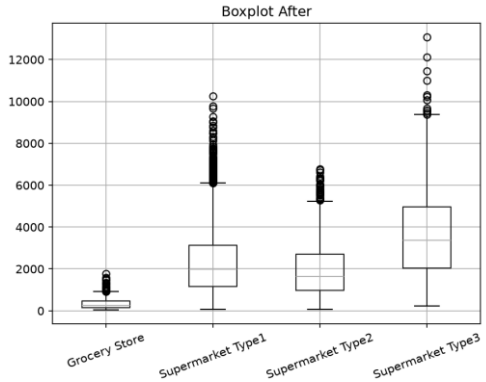
# Boxplot after
data.boxplot(
```

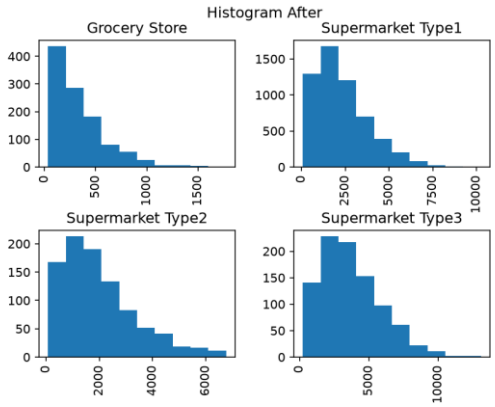
```
    column="Item_Outlet_Sales",
    by="Outlet_Type"
)

plt.title("Boxplot After")
plt.suptitle("")
plt.xlabel("Outlet Type")
plt.ylabel("Item Outlet Sales")
plt.xticks(rotation=20)
plt.show()
# Histogram after
data.hist(
    column="Item_Outlet_Sales",
    by="Outlet_Type",
    bins=10
)
plt.suptitle("Histogram After")
plt.show()
print("\nNumber of bins = 10")
```









```
Data:
  Item_Identifier  ...  Item_Outlet_Sales
0           FDA15  ...          3735.1380
1           DRC01  ...           443.4228
2           FDN15  ...          2097.2700
3           FDX07  ...           732.3800
4           NCD19  ...           994.7052

[5 rows x 12 columns]

Data Types:
Item_Identifier                 str
Item_Weight                 float64
Item_Fat_Content                str
Item_Visibility             float64
Item_Type                       str
Item_MRP                    float64
Outlet_Identifier               str
Outlet_Establishment_Year     int64
Outlet_Size                     str
Outlet_Location_Type            str
Outlet_Type                     str
Item_Outlet_Sales           float64
dtype: object
```

```
Frozen Foods:
     Item_Identifier  ...  Item_Outlet_Sales
8              FDH17  ...          1076.5986
9              FDU28  ...          4710.5350
24             FDR28  ...          4078.0250
51             FDM40  ...           850.8924
63             FDY40  ...           147.8076
...              ...  ...                ...
8477           FDC05  ...          1970.7680
8482           FDZ28  ...           763.0068
8507           FDN28  ...           515.9950
8511           FDF05  ...          4207.8560
8517           FDF53  ...          3608.6360

[856 rows x 12 columns]

After Appending 5 Records:
     Item_Identifier  ...  Item_Outlet_Sales
8523           FDA15  ...          3735.1380
8524           DRC01  ...           443.4228
8525           FDN15  ...          2097.2700
8526           FDX07  ...           732.3800
8527           NCD19  ...           994.7052

[5 rows x 12 columns]
```

```
Data in Ascending Order:
     Item_Identifier  ...  Item_Outlet_Sales
7467           DRA12  ...           850.8924
1197           DRA12  ...          3829.0158
1693           DRA12  ...           992.7078
118            DRA12  ...          2552.6772
8043           DRA12  ...           283.6308
1245           DRA12  ...          2552.6772
6863           DRA24  ...           982.7208
4416           DRA24  ...          4913.6040
4130           DRA24  ...           327.5736
4900           DRA24  ...          3439.5228

[10 rows x 12 columns]

Number of bins = 10

Missing Values Before:
Item_Identifier                0
Item_Weight                 1463
Item_Fat_Content               0
Item_Visibility                0
Item_Type                      0
Item_MRP                       0
Outlet_Identifier              0
Outlet_Establishment_Year      0
Outlet_Size                 2411
Outlet_Location_Type           0
Outlet_Type                    0
Item_Outlet_Sales              0
dtype: int64
```

```
Number of Rows After:
8528

Missing Values After:
Item_Identifier                0
Item_Weight                    0
Item_Fat_Content               0
Item_Visibility                0
Item_Type                      0
Item_MRP                       0
Outlet_Identifier              0
Outlet_Establishment_Year      0
Outlet_Size                    0
Outlet_Location_Type           0
Outlet_Type                    0
Item_Outlet_Sales              0
dtype: int64

Number of bins = 10
```